

# LAB PROGRAMS 27,28,29

## 27. Vertical Bar Chart

### Aim

To write a Python program to display a **bar chart** showing the popularity of different programming languages.

### Algorithm

1. Import the matplotlib.pyplot library.
2. Store the programming languages in a list.
3. Store their popularity values in another list.
4. Use plt.bar() to create the bar chart.
5. Add title and axis labels.
6. Display the chart using plt.show().

### Coding

```
import matplotlib.pyplot as plt

languages = ["Java", "Python", "PHP", "JavaScript", "C#", "C++"]

popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

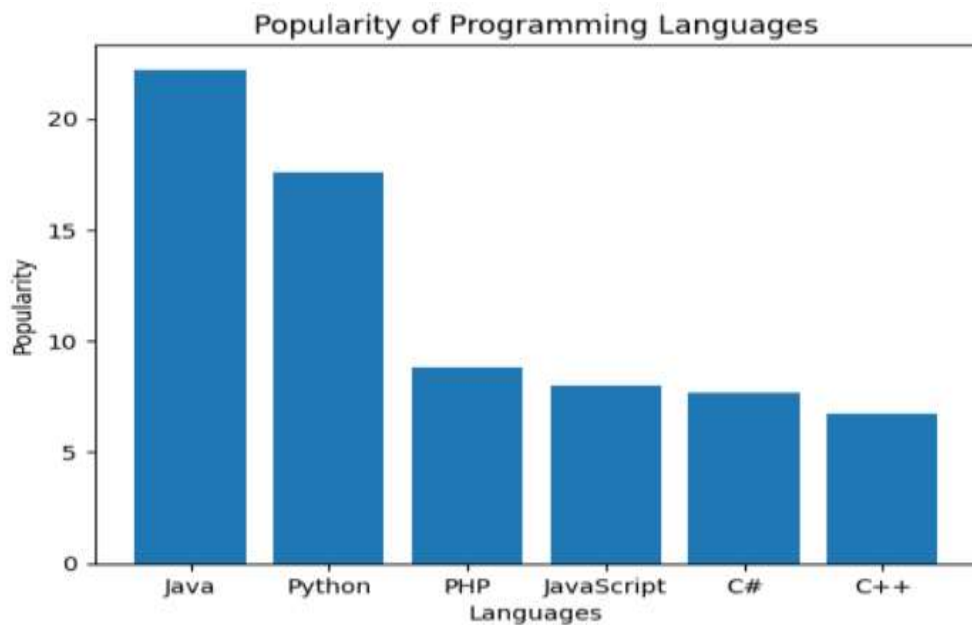
plt.bar(languages, popularity)

plt.title("Popularity of Programming Languages")

plt.xlabel("Languages")

plt.ylabel("Popularity")

plt.show()
```



## 28. Horizontal Bar Chart

### Aim

To write a Python program to display a **horizontal bar chart** showing the popularity of programming languages.

### Algorithm

1. Import the matplotlib.pyplot library.
2. Create lists for programming languages and popularity.
3. Use `plt.barh()` to create a horizontal bar chart.
4. Add the title and axis labels.
5. Display the chart using `plt.show()`.

## Coding

```
import matplotlib.pyplot as plt

languages = ["Java", "Python", "PHP", "JavaScript", "C#", "C++"]

popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

plt.barh(languages, popularity)

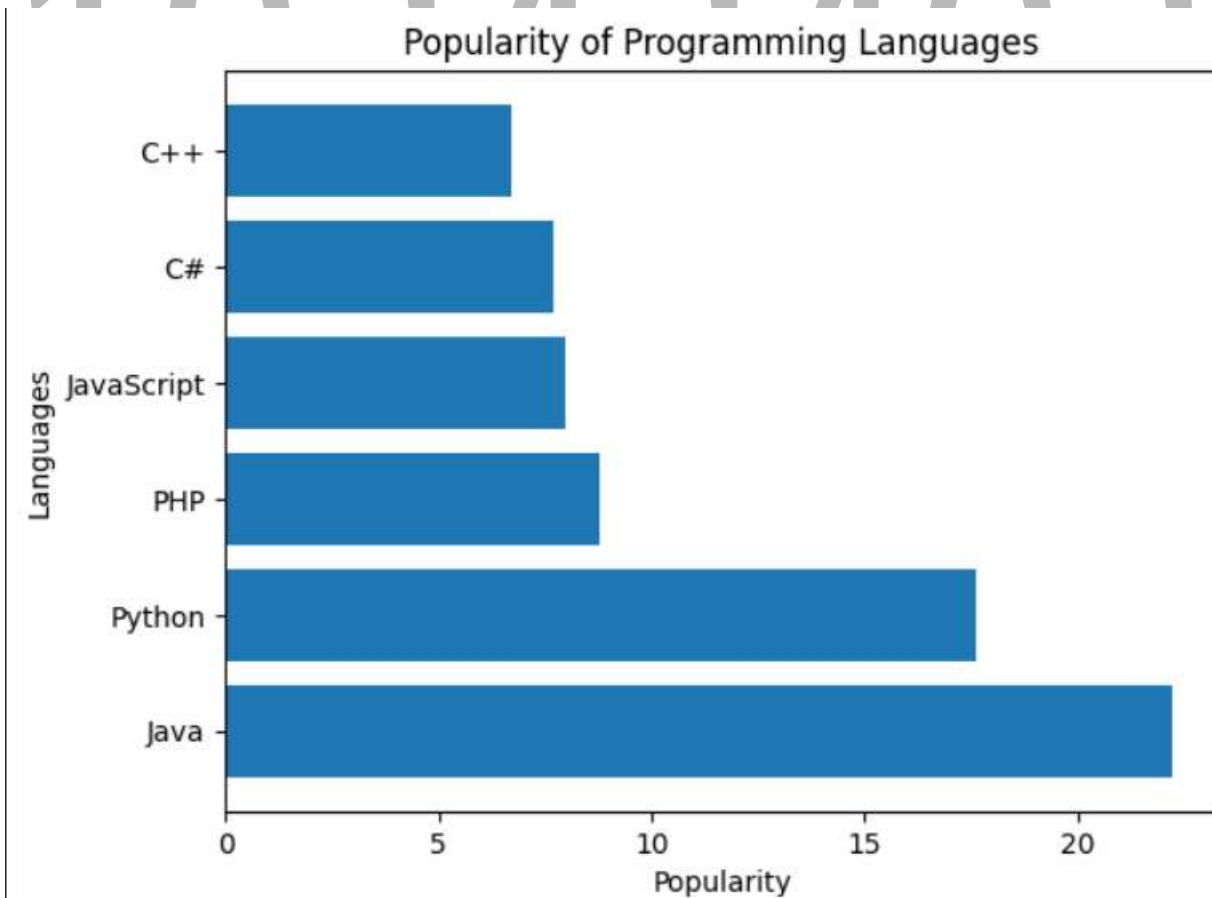
plt.title("Popularity of Programming Languages")

plt.xlabel("Popularity")

plt.ylabel("Languages")

plt.show().
```

## Output:



## 29. Bar Chart with Different Colors

### Aim

To write a Python program to display a **bar chart with different colors** for each programming language.

### Algorithm

1. Import the matplotlib.pyplot library.
2. Store the programming languages and their popularity values.
3. Create a list containing different colors.
4. Use plt.bar() with the color list.
5. Add the title and axis labels.
6. Display the chart using plt.show().

### Coding

```
import matplotlib.pyplot as plt

languages = ["Java", "Python", "PHP", "JavaScript", "C#", "C++"]

popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

colors = ["red", "black", "green", "blue", "yellow", "cyan"]

plt.bar(languages, popularity, color=colors)

plt.title("Popularity of Programming Languages")

plt.xlabel("Languages")

plt.ylabel("Popularity")

plt.show()
```

Output:

